ALAN ANDRE POMBOZA

Full-Stack Developer · AI Engineer | .NET, Angular, Python, LLMs

St. Catharines, ON | alan4630@gmail.com

linkedin.com/in/alan-andre-pomboza | github.com/AlanJuker | alanjuker.github.io

PROFESSIONAL SUMMARY

Full-stack developer and AI engineer with 3 years of experience shipping production systems in C#/.NET, Angular, Oracle and Python. Built national election systems for Ecuador's National Electoral Council, including a machine learning and computer vision identity-verification module. Built Databricks pipelines and AI solutions with Meta Llama models for a US client. Completing a Master of Computer Science in Applied AI. Bilingual: English (C1 certified) and Spanish (native).

TECHNICAL SKILLS

- **Programming:** Python, C#, TypeScript, JavaScript, SQL, PL/SQL
- **AI / Machine Learning:** LLM applications (Claude, OpenAI, Meta Llama, structured outputs), embeddings, hybrid retrieval, scikit-learn, computer vision (liveness and spoof detection, TensorFlow.js), MLflow, model evaluation, agentic AI
- **Back-end:** .NET 8, .NET 6, ASP.NET Core Web API, .NET Framework 4.7, REST APIs, FastAPI, SQLAlchemy
- **Front-end:** Angular 16, React, HTML, CSS
- **Data:** Databricks, Oracle 19c (stored procedures), PostgreSQL, SQL Server, MySQL, Power BI, XtraReports
- **DevOps & Tools:** Azure DevOps (CI/CD, Repos, Boards), GitHub Actions, Docker, Git, Postman, Scrum / Agile

PROFESSIONAL EXPERIENCE

Full-Stack Developer — **Sep 2023 – Oct 2026**

National Electoral Council (CNE) — *Quito, Ecuador*

- Automated political-organization financing accounting, previously a manual process, by building the full system: Oracle 19c stored procedures, .NET 6 / 8 Web APIs and Angular user interfaces.
- Built the candidacy simulator used for all candidacies to automatically check legal gender-parity and youth-inclusion requirements.
- Built Voto Informado, the public web application where voters review candidates' work plans and proposals.
- Built the online electoral address-change system and its machine learning and computer vision anti-spoofing module (person vs. spoof detection), which verifies citizen identity for automated requests.
- Delivered executive reports in XtraReports and Power BI; shipped releases through Azure DevOps CI/CD pipelines.

AI Software Engineer (Contract) — **Dec 2025 – Feb 2026**

Stimulus, Inc. — *Remote, United States*

- Built Databricks data pipelines in Python and SQL and production AI solutions with Meta Llama and MLflow for a US client.
- One of three engineers mentored by a Senior AI/ML Engineer specializing in agentic AI, from requirements to delivery.

PROJECTS

noc-noc – NOC 2021 occupation-code finder — github.com/AlanJuker/noc-noc

- Hybrid TF-IDF and embedding retrieval (reciprocal rank fusion) puts the right code in the top 5 for 93.6% of 5,468 held-out official titles and 79.4% of 4,242 real Job Bank titles. Python, scikit-learn, embeddings, Claude API, FastAPI

signal-problems – Toronto subway delay analytics — github.com/AlanJuker/signal-problems

- Resolved 99.7% of 260,000 TTC delay records (1,251 station spellings to 70 stations) without guessing; serves an API, an interactive subway map and a commute-risk check. Python, FastAPI, PostgreSQL, React, TypeScript, Docker, GitHub Actions

EDUCATION

Master of Computer Science in Applied Artificial Intelligence · *University of Niagara Falls Canada* — **2025 – Present**

Master's Degree in Software Engineering and Computer Systems · *International University of La Rioja (UNIR)* — **2023 – 2024**

Bachelor's Degree in Systems Engineering · *Pontifical Catholic University of Ecuador (PUCE)* — **2019 – 2023**

CERTIFICATIONS

AI Skills Accelerator – Awana (2026) · GitHub Foundations (2025) · Postman API Fundamentals (2024) · Python Programming – UNIR (2024) · Microsoft Azure Fundamentals AZ-900 (2023) · Scrum Foundation Professional (SFPC) – CertiProf (2023) · Application & Database Security – UNAM (2021)